

Supporting Information: Estimand Divergence: Static Docking Versus Dynamics for Antimalarial Triage

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Table S1: Evidence map. Status of each supporting analysis: PASS = meets its pre-declared criterion; descriptive = reported without a pass/fail gate; null = no signal detected.

Analysis	Table	Status
Docking validation (MMV consensus)	S1	PASS (ROC-AUC 0.924–1.000)
DEKOIS 2.0 PfDHFR (Vina alone)	S1	Null (ROC-AUC 0.450, chance level)
Homology-model QC (6 mutants)	S-Homology-QC	PASS (all thresholds met)
RRS threshold sensitivity	S11	Descriptive (Class D 1 → 9)
GNINA rescoring (39-panel)	S15	Descriptive (agreement near-degenerate)
ChEMBL fold-shift comparison	S20	Null (ρ -0.13–0.19)
PP-01 biophysical audit	S19	PASS (protocol-local)
K76A replicate sensitivity	S19	No claim (7.75 kcal/mol)
Parent-MD ligand identity	archived review	Documented (no consensus-lead status)

Table S2: Docking enrichment and re-docking accuracy. Values are from corrected parent-study benchmarks using V2 grids [1]. MMV Malaria Box values are consensus-score ROC-AUCs for PfDHFR, PfCRT, and PfATP4; DEKOIS 2.0 used AutoDock Vina for PfDHFR. BEDROC values use $\alpha = 20$. Re-docking RMSD is the heavy-atom RMSD between the top-scoring pose and the co-crystallized ligand; the MTX RMSD was approximately 30 Å.

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC	RMSD (Å)
PfDHFR (7F3Y)	Vina redock	–	–	–	–	N/A
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309	–
	DEKOIS (Vina only)	0.450	–	0.00	–	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434	–
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–
PfClpR (4GM2; not PfClpP)	MMV enrichment	not evaluated	–	–	–	–

Target identity. PDB 4GM2 is PfClpR rather than PfClpP; it is not used as PfClpP validation.

State-specific summaries. For PfDHFR, mean RRS values were 74.7 (N51I), 73.7 (C59R), 75.3 (S108N), and 76.8 (I164L). For PfCRT, the corresponding means were 85.4 (K76T) and 87.0 (K76A). These target-specific summaries describe score retention and do not measure biochemical resistance.

Varying the wild-type eligibility threshold from 4.0 kcal mol⁻¹ to 5.0 kcal mol⁻¹ left class counts unchanged. Raising retention thresholds from 70 %/80 % to 80 %/90 % increased Class D from 1 to 9. These cut-offs are operational triage rules for this panel, not calibrated resistance categories.

Secondary calibration and ranking analyses

Tartarus docking calibration

The Tartarus analysis used QuickVina[2], an AutoDock Vina[3] derivative, on 19 913 primary leads against PDB entries 1SYH, 6Y2F, and 4LDE. The composite MPO comparison included

Table S3: Homology-model quality for the six resistance-associated states. Models were built with SWISS-MODEL from the indicated wild-type templates and retained only if QMEAN > -4.0, GMQE > 0.7, Ramachandran favored residues > 95 %, and C α RMSD to the template < 1.5 Å. Detailed per-model scores are kept with the SWISS-MODEL outputs; the table records whether each model passed all four thresholds.

Target	Mutant state	WT template (PDB)	QC gate
PfDHFR	N51I	7F3Y	PASS
PfDHFR	C59R	7F3Y	PASS
PfDHFR	S108N	7F3Y	PASS
PfDHFR	I164L	7F3Y	PASS
PfCRT	K76T	6UKJ	PASS
PfCRT	K76A	6UKJ	PASS

All six models were used as docking receptors. Passing these thresholds does not imply experimental structural accuracy.

Independent P2Rank pocket audit vs P2 Vina docking boxes

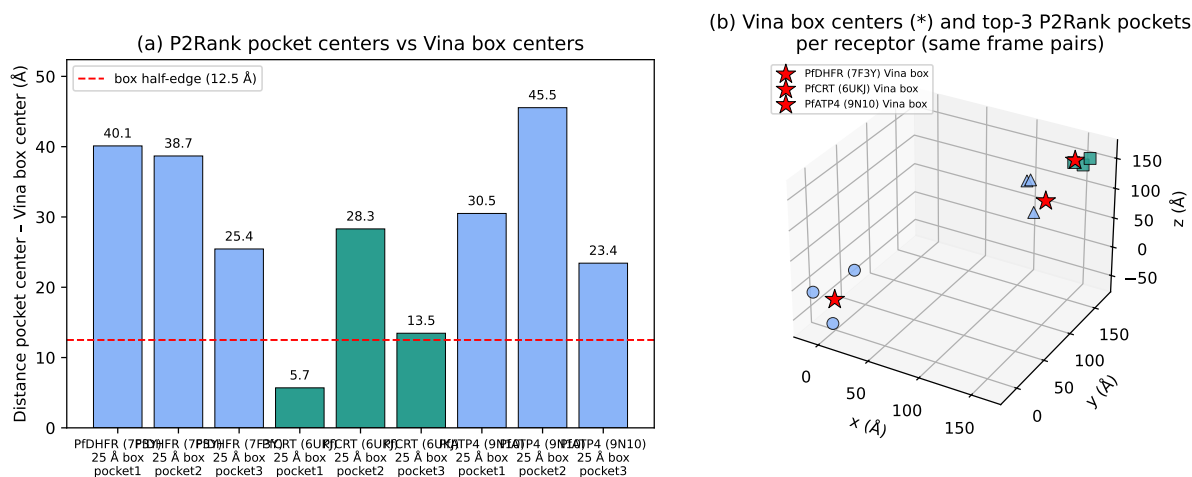


Figure S1: P2Rank pocket prediction vs. Vina grid-box alignment. Top P2Rank pocket probabilities compared with the fixed Vina grid-box centers: PfDHFR (7F3Y, prob 0.972; top pockets 25.5–40.1 Å from the box center), PfCRT (6UKJ, prob 0.999; top pocket 5.7 Å from the box center — the sole concordant case, corroborating the docking box), PfATP4 (9N10, prob 0.663; top pockets 23–46 Å away, a frame/provenance caveat rather than a refutation), and PfClpP (2F6I, prob 0.299; no Vina grid exists for this receptor). Left panel: top-3 pocket-to-box-center distances; 6UKJ highlighted as concordant. Source: P2Rank 2.5.1 predictions (no volume field is present in the artifact).

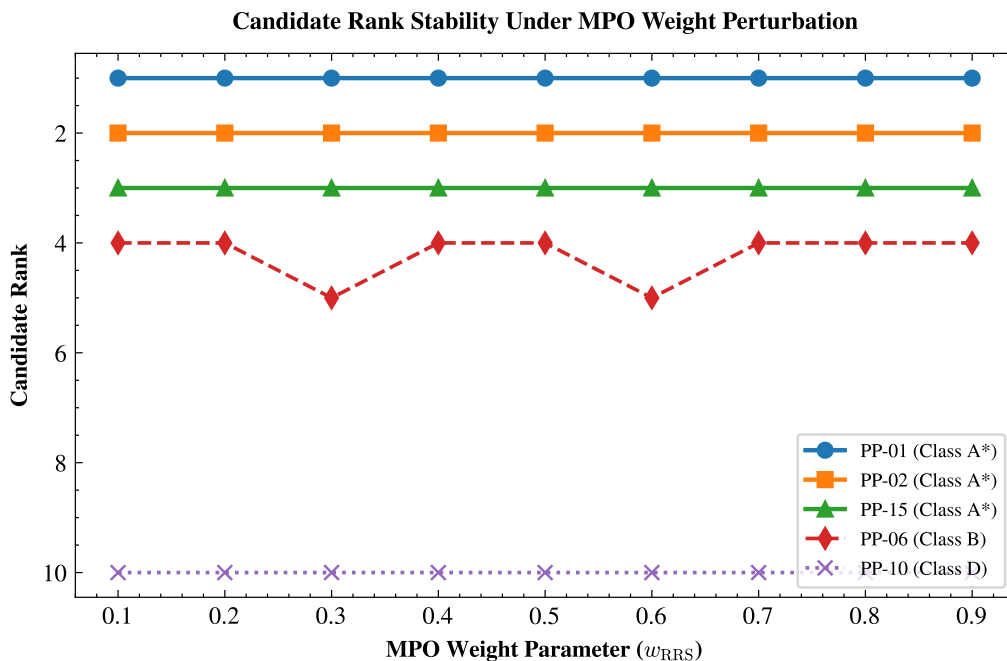


Figure S2: *MPO rank invariance under weight perturbation.* Candidate rank trajectories across composite MPO weight variations ($w_{\text{RRS}} \in [0.1, 0.9]$).

Table S4: *Predicted ADMET profile of the 17 set-C polypharmacology candidates (ADMET-AI, parent-library screening).* All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Across candidates, LogS ranged from 0.34 to 0.73; CYP3A4 inhibition was ≥ 0.5 for 7/17 candidates, Caco-2 permeability was ≥ 0.5 for 15/17, and Clint_h ranged from 0.48 to 0.63.

Compound	ADMET-AI LogS	CYP3A4 prob.	Caco-2 prob.	Clint_h
PP-01	0.34	0.39	0.63	0.60
PP-02	0.46	0.97	0.86	0.55
PP-03	0.43	0.62	0.95	0.55
PP-04	0.40	0.58	0.92	0.63
PP-05	0.61	1.00	0.35	0.51
PP-06	0.54	0.90	0.97	0.60
PP-07	0.36	0.02	0.75	0.62
PP-08	0.59	1.00	0.41	0.56
PP-09	0.65	0.13	0.81	0.57
PP-10	0.49	0.13	0.93	0.62
PP-11	0.39	0.06	0.96	0.58
PP-12	0.49	0.00	0.98	0.48
PP-13	0.57	0.97	0.98	0.53
PP-14	0.66	0.05	0.68	0.49
PP-15	0.66	0.49	0.95	0.51
PP-16	0.54	0.25	0.91	0.50
PP-17	0.73	0.01	0.82	0.50

Table S5: Candidate-cohort ACSI weight sensitivity. One ACSI component weight was varied by $\pm 20\%$ and the remaining weights were rescaled proportionally to preserve a unit sum. Mean and standard deviation refer to the resulting ACSI values across the 17 candidates. Rank stability is measured by Spearman correlation over all candidates and Jaccard overlap of the top five candidates with the baseline ranking. Because the full-library component matrix was unavailable, this analysis applies only to the 17-candidate Set-C cohort; it does not assess the full-library top-20 ranking.

Weight varied	Variation (%)	New weight	Mean ACSI	SD	Spearman ρ	Top-5 Jaccard
D_{DrugBank}	-20	0.320	0.512	0.149	0.977 9	0.666 7
D_{DrugBank}	+20	0.480	0.574	0.159	0.933 8	0.666 7
D_{ANPDB}	-20	0.200	0.568	0.149	0.916 7	0.666 7
D_{ANPDB}	+20	0.300	0.518	0.158	0.980 4	0.666 7
f_{sp^3}	-20	0.160	0.552	0.152	0.951 0	0.666 7
f_{sp^3}	+20	0.240	0.534	0.153	0.958 3	0.666 7
NPL	-20	0.120	0.535	0.160	0.963 2	0.666 7
NPL	+20	0.180	0.551	0.145	0.980 4	1.000 0

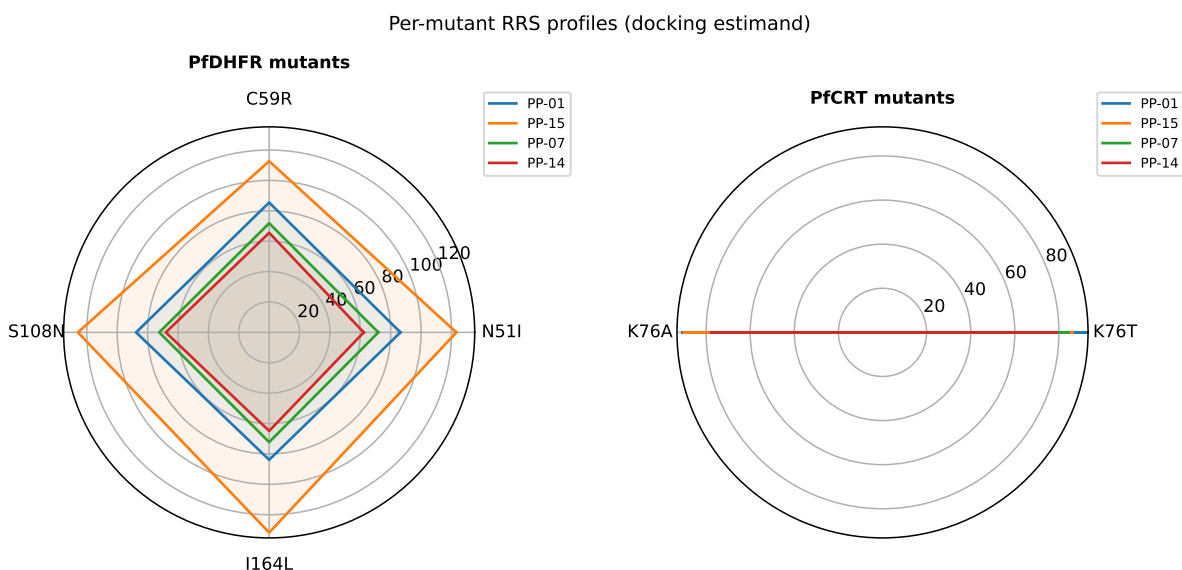


Figure S3: Per-mutant RRS profiles. Docking RRS per mutant state for representative complete-two-target candidates (PP-01, A*; PP-15, A; PP-07, B; PP-14, D) on the PfDHFR (N51I/C59R/S108N/I164L) and PfCRT (K76T/K76A) channels, from the RRS classification records.

Table S6: Sensitivity of the available-target RRS classes to the WT eligibility and retention thresholds. Counts are reconstructed from the versioned threshold-sensitivity output for all 17 candidates. Thresholds used in the main analysis are shown in bold. The table checks classification stability only; it does not calibrate cut-offs against experimental resistance.

WT threshold	Lower retention	Upper retention	A*	A	B	C	D
5.0	70	80	5	1	5	5	1
5.0	75	85	3	1	4	3	6
5.0	80	90	2	0	4	2	9
4.0	70	80	5	1	5	5	1
6.0	70	80	6	0	5	5	1
7.0	70	80	6	0	5	5	1
7.0	75	85	4	0	4	3	6
7.0	80	90	2	0	4	2	9

Table S7: Target-specific docking-score retention in the eligible candidates. Values are summarized across candidates with an eligible WT denominator for the indicated target. The target-specific class counts are descriptive and are not equivalent to the primary two-target classification.

Target	Eligible candidates	Mean RRS	Median RRS	Range	Class A*/A	Class B/C/D
PfDHFR	12	75.1	70.8	59.7–123.4	2	10
PfCRT	17	86.2	83.5	74.7–104.8	10	7

Table S8: Questions addressed and interpretation boundaries. The Set-C docking cohort has 17 candidates and 136 wild-type/mutant systems after prioritization filters; 12 candidates have complete PfDHFR/PfCRT panels and define the primary RRS analysis, and 5 are PfCRT-only records reported separately. Parent-study MD systems do not overlap Set-C. The 16-system Set-C pilot passed geometric trajectory checks and is analyzed separately; its MD-RRS and MM-GBSA values are not used to assign docking-RRS classes. Docking-RRS is a computational triage estimand, whereas MD-RRS and MM-GBSA are secondary structural and endpoint diagnostics.

Evidence layer	Cohort	Systems	Primary analyses	Interpretation boundary
Set-C docking	PP-01–PP-17	136 (12 complete; 5 PfCRT-only)	WT/mutant docking; target-specific RRS, ACSI, PNS	Primary computational triage cohort; not a measure of affinity, engagement, or biological resistance
Parent-study MD	Ligands 164, 201, 214, 438	4 WT complexes × 10 ns	RMSD, contacts, bound-state assessment, MM-GBSA where valid	Structural check; one interpretable MM-GBSA estimate (PfCRT–214)
Set-C MD pilot	PP-01/PP-02	16 systems (all met trajectory criteria)	Trajectory geometry, MD-RRS, MM-GBSA	Single-replicate structural stress test; non-equivalent to docking-RRS and separate from its classification
Full Set-C MD-RRS	PP-01–PP-17	136 planned systems	Full trajectory QC and MD-RRS	Not available because the complete cohort was not simulated

Table S9: Sensitivity of the PNS ranking to the imputed PfCRT centrality. The reference value is the observed network-mean imputation ($C_{\text{PfCRT}} = 0.151$). Alternative values were zero, one-half, one-and-a-half, and twice the reference value. Spearman ρ is calculated against the reference 17-candidate ranking; all candidates have two target entries in the PNS record. The ranking is sensitive to this network-data choice; target essentiality and biological polypharmacology were not assessed.

Imputation	C_{PfCRT}	Spearman ρ	Minimum PNS	Maximum PNS	n
Zero	0.000 0	0.971 6	0.300	5.147	17
One-half canonical	0.075 5	0.997 5	0.674	5.573	17
Canonical network mean	0.151 0	1.000 0	1.047	6.000	17
One-and-a-half canonical	0.226 5	0.975 5	1.421	6.426	17
Twice canonical	0.302 0	0.963 2	1.795	6.853	17

up to 17 211 complete observations. The composite association was $\rho = 0.0129$ ($p = 0.0913$); target-specific associations were small and do not imply score equivalence or independence.

Table S10: *Tartarus docking calibration: Spearman correlations between QuickVina scores and weighted MPO scores.*

Score	Spearman ρ	p -value	n	Interpretation
Composite (mean of 3 targets)	0.0129	9.13×10^{-2}	17 211	No meaningful association
1SYH (PfDHFR)	0.0655	8.18×10^{-18}	17 205	Small effect
6Y2F (PfCRT)	-0.0040	6.04×10^{-1}	17 211	Null
4LDE (A2A receptor)	-0.1415	1.17×10^{-77}	17 211	Small effect

Tartarus cross-docking correlation

Only 24 molecules overlapped between the Tartarus and project docking records. Because the structures were not matched, this comparison cannot calibrate the docking scores experimentally.

Table S11: *Tartarus independent-tool comparison for the 24-molecule overlap.*

Tartarus	Project metric	n	ρ	p	Interpretation
Composite	4GM2/PfClpR	24	0.246	2.47×10^{-1}	No
Composite	6UKJ/PfCRT	24	0.075	7.28×10^{-1}	No
Composite	7F3Y/PfDHFR	24	-0.110	6.10×10^{-1}	No
Composite	9N10/PfATP4	24	0.118	5.83×10^{-1}	No
1SYH/DHFR	7F3Y/PfDHFR	24	-0.442	3.04×10^{-2}	Exploratory

Complete ACSI components

ACSI was calculated as

$$\text{ACSI} = 0.40D_{\text{DrugBank}} + 0.25D_{\text{ANPDB}} + 0.20f_{sp^3} + 0.15\text{NPL}.$$

Components were min-max normalized over the 17-candidate cohort. Two candidates exceeded $\text{ACSI} > 0.70$, and the cohort mean was 0.543. Varying one weight at a time by $\pm 20\%$ gave Spearman correlations of 0.9167–0.9804 and top-five Jaccard overlaps of 0.6667–1.0000.

PNS ranking and imputation

The STRING[4] network used confidence threshold 700; PfCRT centrality was imputed with the network mean, 0.151. The full ranking is listed in the table. Replacing the imputed value with zero, one-half, one-and-a-half, or twice the reference value gave rank correlations of 0.9632–1.0000. This result reflects sensitivity to a network-data choice, not biological target importance.

Predicted ADMET

DiffDock-Vina and activity-model comparisons

In the 65 856-molecule library, Vina[3] scores and DiffDock[5] confidence were only weakly related overall. A Benjamini-Hochberg analysis of 18 pairwise activity comparisons across compound

Table S12: *PNS ranking and RRS class. PfCRT-only denotes candidates without an eligible PfdHFR WT denominator for primary RRS.*

Candidate	PNS	ACSI	RRS class
Rank 1	6.00	0.599	C
Rank 2	4.71	0.308	C
Rank 3	4.53	0.567	C
Rank 4	4.48	0.603	B
Rank 5	4.48	0.551	B
Rank 6	4.45	0.173	B
Rank 7	4.45	0.459	A*
Rank 8	4.36	0.743	D
Rank 9	4.36	0.605	C
Rank 10	4.34	0.594	C
Rank 11	4.20	0.601	B
Rank 12	3.50	0.576	A
Rank 13	2.32	0.597	PfCRT-only
Rank 14	1.23	0.823	PfCRT-only
Rank 15	1.14	0.387	PfCRT-only
Rank 16	1.10	0.458	PfCRT-only
Rank 17	1.04	0.589	PfCRT-only

groups using three Ersilia[6] models found no adjusted-significant differences; all rank-biserial effect sizes were negligible. These analyses provide no evidence of biological activity.

The ADMET-AI[7] endpoint table for all 17 candidates is given in Table S4. These predictions were not experimentally validated or compared across tools.

Target identifiers were taken from PlasmoDB/VEuPathDB (accessed 12 August 2026) [8]; the release string was not recorded. A parent-study PfCRT redocking check at pH 5.2 (100/100 ligands; $\rho = 0.270$, $p = 0.006551$) was not merged into the Set-C cohort. No GO or pathway enrichment was run, as no multi-gene targetome or background was defined [9].

Parent-study MD (non-overlapping cohort)

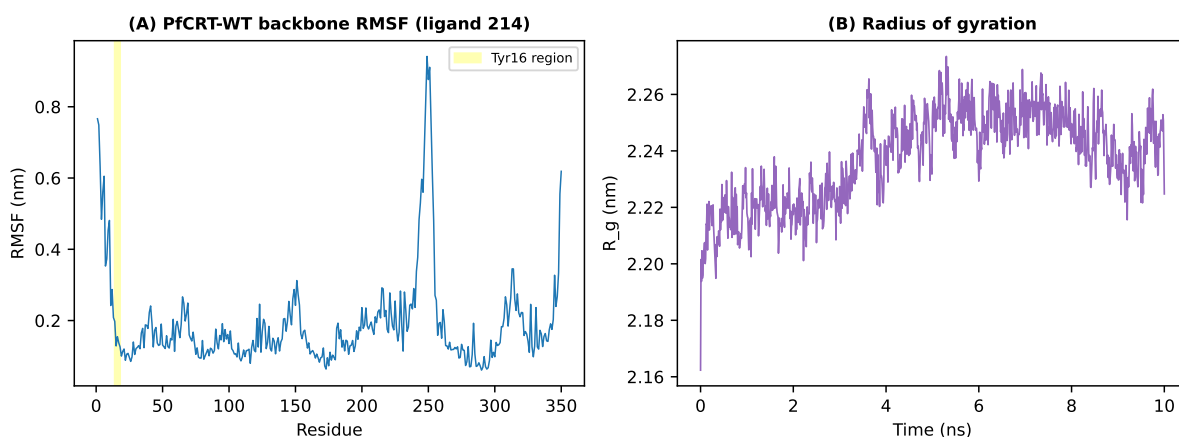


Figure S4: *Parent-study PfCRT-WT (system 214) diagnostics. (A) Per-residue backbone RMSF from the canonical 10 ns trajectory (gmæ rmsf; the Tyr16 anchor region is retained in the shaded band). (B) Backbone radius of gyration over the production run. Source: parent-study production trajectory analyses.*

Table S13: Evidence scope and next validation. The computational results support prioritization and structural hypotheses within the stated cohorts; they do not substitute for biochemical, biophysical, phenotypic, or clinical measurements. The table separates the primary computational triage estimand from secondary diagnostics, technical checks, and unobserved biological quantities.

Evidence layer	Supported interpretation	Next validation needed
Docking-RRS	Relative retention of empirical docking scores across the selected mutant panel; primary computational triage only	Matched biochemical activity measurements for wild-type and mutant PfDHFR/PfCRT
ACSI/PNS	Cohort-relative chemical-space and network-weighted ranking dimensions	Larger, independently selected panels and experimentally anchored target/network data
Parent-study MD	Ligand retention under the declared geometric criterion and system-specific endpoint assessment	Replicated, membrane-aware simulations and orthogonal binding or transport assays
Set-C MD pilot	Secondary trajectory-derived structural stress test for PP-01 and PP-02; single replicate per system and non-equivalent to docking-RRS	Independent mutant/wild-type replicates with declared convergence and uncertainty analyses
Biological polypharmacology	Not measured by the present computational scores	Target engagement plus parasite-phenotype and mechanism-of-action experiments

Table S14: Scope of the targeted resistance panel. The panel was designed to test resistance-aware ranking across selected molecular contexts; it was not designed to estimate resistance prevalence, reconstruct the parasite targetome, or establish pathway-level polypharmacology.

Dimension	Covered in this study	Not covered
PfDHFR	WT plus N51I, C59R, S108N, and I164L docking states	Other resistance haplotypes, prevalence, and experimental activity
PfCRT	WT plus K76T and K76A docking states; neutral-pH protocol	Full haplotype diversity, acidic-vacuole conditions, and transport measurements
PfATP4	One parent-study structural context	Mutant-state panel and resistance inference
PfClpR	One parent-study context associated with PDB 4GM2	PfClpP validation and broader proteostasis target coverage
Chemical cohort	17 filtered Set-C candidates; 12 define the complete two-target primary analysis	Representative sampling of the 65,856-molecule library
Biological outcome	Computational docking, trajectory, and endpoint summaries	Direct target engagement, parasite phenotype, and pathway-level mechanism

Table S15: Stable *Plasmodium falciparum* target identifiers and mutation contexts. The identifiers define targets and states; they are not pathway-level or experimental evidence.

Target ID	Context	Mutation / boundary
PfDHFR PF3D7_0417200	7F3Y WT template	N51I/C59R/S108N/I164L; antifolate states
PfCRT PF3D7_0709000	6UKJ WT template	K76T resistance-associated state; K76A mechanistic comparator
PfATP4 PF3D7_1211900	9N10 parent-study structure	Annotation only; outside Set-C docking panel
PfClpP PF3D7_0307400	Proteolytic Clp subunit	Not assigned to 4GM2; no PfClpP claim
PfClpR PF3D7_1436800	4GM2	4GM2 is PfClpR, not PfClpP

Table S16: *MM-GBSA for the four parent-study complexes. These systems are not among the 17 Set-C candidates. System 214 (PfCRT-WT) is reported as mean $\pm$ SEM over 101 snapshots; dissociated or conversion-invalid systems are N/A.*

System	Target	ΔG_{bind}	SEM	Interpretation
System 164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated [‡]
System 201	PfDHFR-WT	–	–	N/A; ligand dissociated [‡]
System 214	PfCRT-WT	–18.25	0.40	Interpretable (gmx_MMPBSA [10])
System 438	PfATP4-WT	–	–	N/A; conversion corrupted [§]

*PDB 4GM2 is PfClpR, not PfClpP.

[‡]Ligand left the site during production; no binding free energy is inferred.

[§]Two-chain topology incompatible with CHARMM36[11]-to-AMBER conversion.

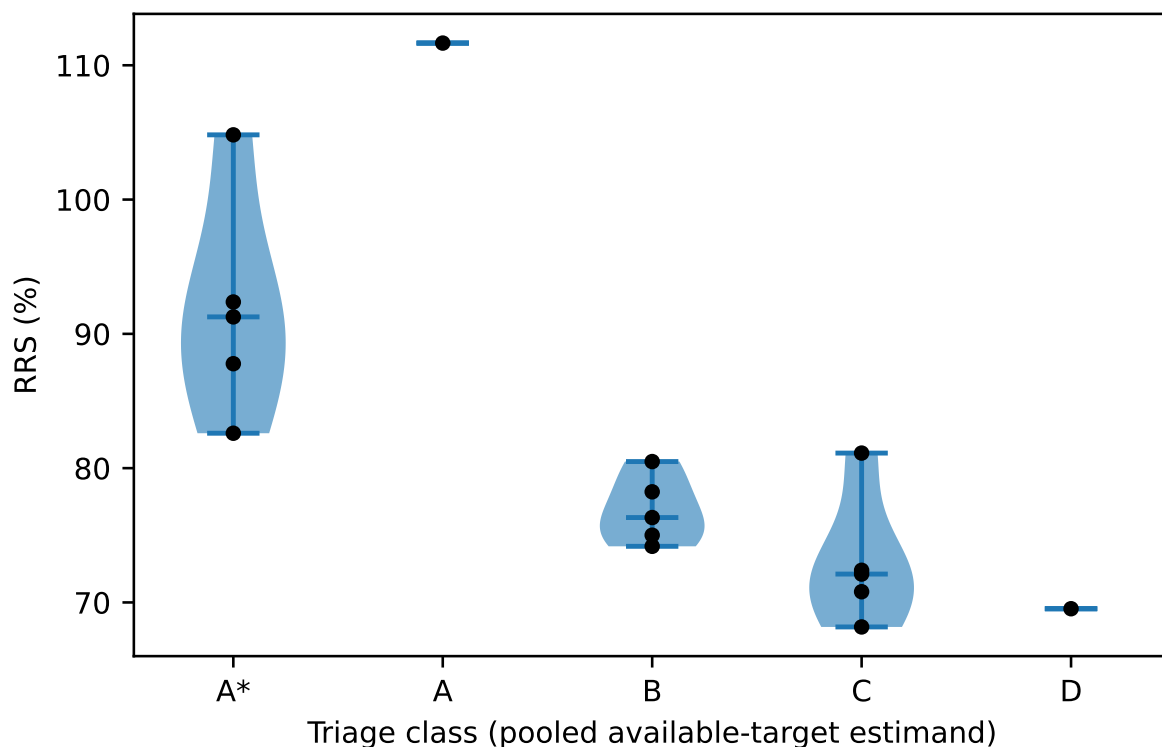


Figure S5: *RRS distribution across triage classes. Violin plot of per-candidate RRS (pooled available-target estimand) for Classes A* (n = 5), A (n = 1), B (n = 5), C (n = 5), and D (n = 1); black points show individual candidates.*

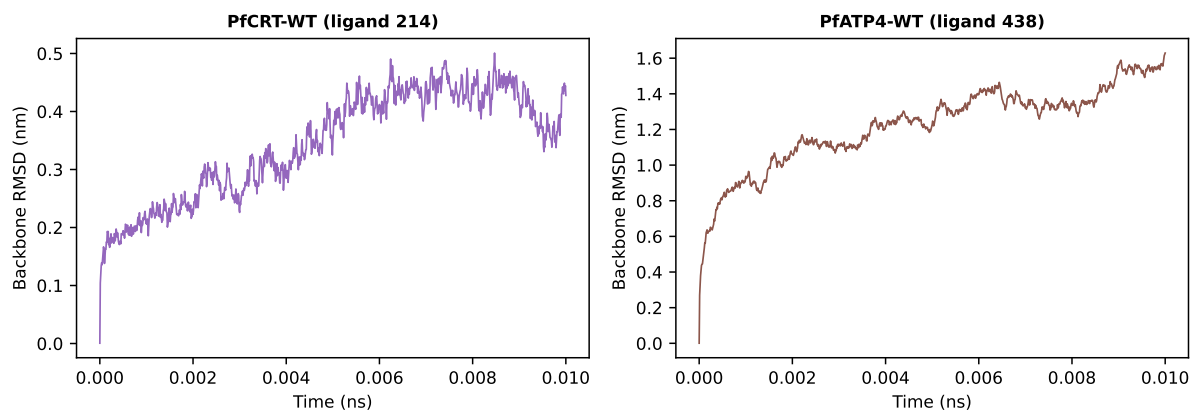


Figure S6: *Parent-study PfCRT-WT (system 214) backbone RMSD.* Backbone RMSD over the 10 ns production run (mean 0.34 nm, final-window 0.39 nm); the complex remains stable. Source: parent-study production trajectory analysis.

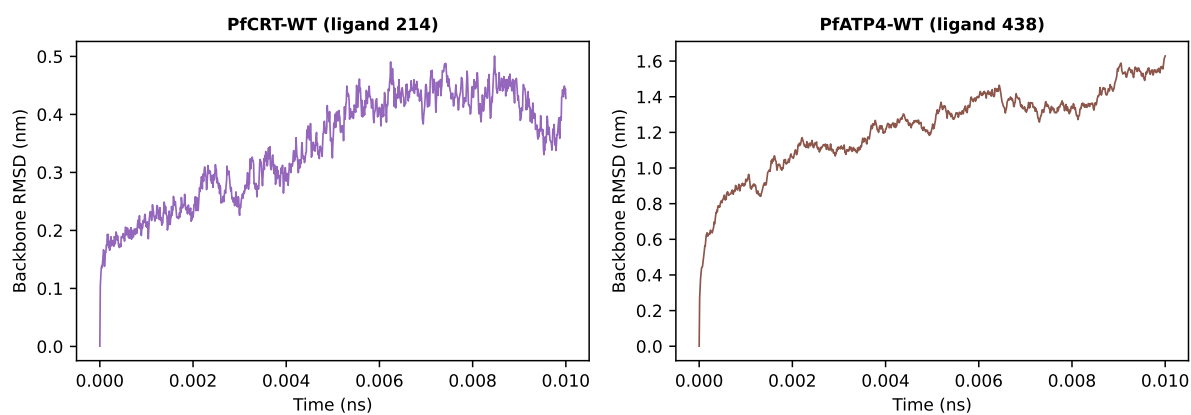


Figure S7: *Parent-study PfATP4-WT (system 438) backbone RMSD.* Backbone RMSD drifts from ~ 0.3 nm to ~ 1.5 nm over the 10 ns run, consistent with the documented force-field conversion corruption of the multi-chain membrane transporter (no interpretable MM-GBSA endpoint is claimed for this system). Source: parent-study production trajectory analysis.

The 16-system single-replicate Set-C pilot on PP-01 and PP-02 tested whether short trajectories and MM-GBSA recover the docking-RRS pattern. All trajectories met geometric acceptance checks. Trajectory metrics and docking-RRS showed opposite operational directions in seven of eight matched mutant comparisons, indicating non-equivalent estimands rather than interchangeable measures of one quantity. Scope boundaries are in [Tables S13 and S14](#).

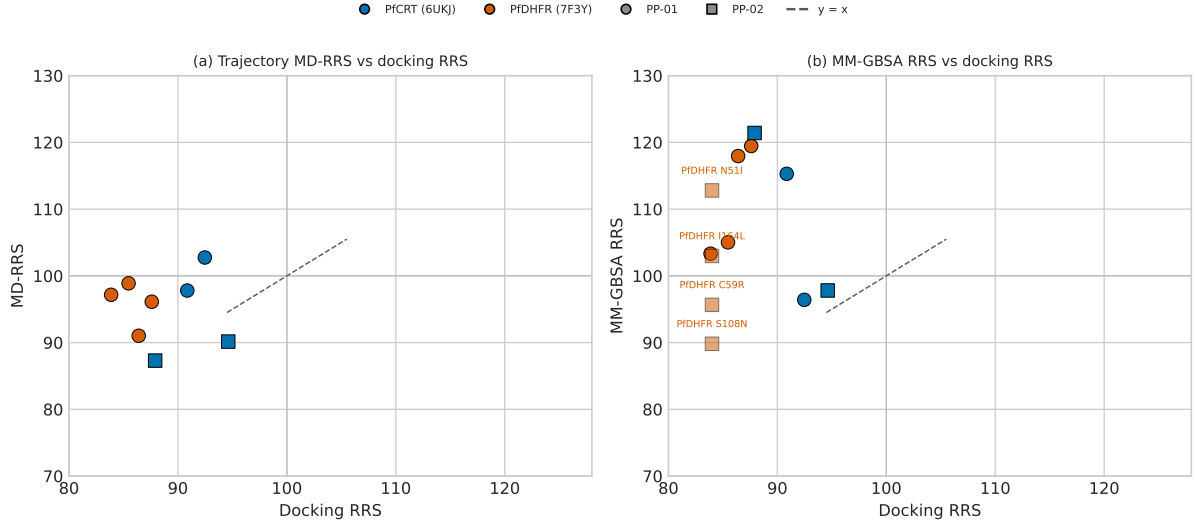


Figure S8: Trajectory-derived ratios versus docking RRS (wild type = 100). (a) Mean minimum-distance ratios for eight mutants with a docking wild-type reference. (b) MM-GBSA ratios versus docking RRS (four PP-02 PfDHFR states lack a docking wild-type reference). The dashed diagonal indicates equality; horizontal and vertical guides denote the wild-type ratio. Neither quantity is a free-energy or experimental resistance measure.

Table S17: Set-C pilot MM-GBSA estimates. ΔG_{bind} is the mean over 100 snapshots $\pm SD_{\text{prop}}$ (within-trajectory SD, not SEM or replicate-level uncertainty). MMG-RRS is the mutant/wild-type endpoint ratio ($\times 100$; wild type = 100). MD-RRS is the corresponding mean minimum-distance ratio. Docking RRS is n/a for PP-02 PfDHFR (no docking wild-type reference). Two systems needed periodic-boundary reconstruction before endpoint analysis. Ratios above 100% indicate retention within endpoint variability, not increased affinity.

Candidate	Target	State	ΔG_{bind} (kcal mol ⁻¹)	SD_{prop}	MMG-RRS	MD-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	97.8	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	102.8	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	100.0	—
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	98.9	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	97.2	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	91.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	96.1	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	100.0	—
PP-02	PfCRT	K76A*	-24.53	1.27	97.8	90.2	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.3	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	100.0	—
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	102.0	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	97.4	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	72.2	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	97.5	n/a
PP-02	PfDHFR	WT*	-30.16	2.06	100.0	100.0	—

Convergence and interaction analysis

The canonical Set-C pilot comprises 16 MM-GBSA endpoints: 12 mutant systems and four WT baselines, each with one production trajectory and 100 stored snapshots. SD_{prop} is the within-trajectory SD; SEM is frame-level only. Neither quantifies between-replicate uncertainty.

Table S18: Endpoint replicate consistency for the PP-01 PfCRT wild-type pillar. Two independent 10 ns productions executed under the identical MDP and analysis chain (GB^{OBC2} igb=5, 0.15 M salt, 100 snapshots) are compared. Tabulated spreads are the within-trajectory SD and the SEM over stored snapshots; the manuscript-wide quoted uncertainty elsewhere is the SEM. The SEM is informational here because MD snapshots are temporally autocorrelated. The between-replicate offset ($|\Delta| = 2.01 \text{ kcal mol}^{-1}$) is reported as an empirical within-protocol endpoint diagnostic; it is not a universal noise floor and is not applied to MD distance ratios. Because the snapshots are temporally autocorrelated, the SEM does not quantify replicate-level uncertainty.

Replicate	Production	ΔG_{bind} (kcal mol $^{-1}$)	SD _{prop}	SEM
R1	production run 1	−30.61	4.12	0.41
R2	independent production run	−28.60	4.35	0.43

The WT replicate is a diagnostic control, not part of the 16-system table; the K76A diagnostic replicate is reported separately below.

Table S19: PP-01 PfCRT K76A inter-replicate endpoint sensitivity. The canonical endpoint and a second reportable trajectory are shown with their propagated within-trajectory SD and frame-level SEM. The difference is a diagnostic measure of inter-replicate sensitivity, not a confidence interval for affinity or a biological resistance effect.

Trajectory	Status	ΔG_{bind} (kcal mol $^{-1}$)	SD _{prop}	SEM	Frames
Canonical	Reportable; 2026-08-19	−35.29	1.17	0.50	100
Second (diagnostic)	Reportable; PBC-whole	−27.54	1.89	0.28	100

Absolute inter-replicate difference: 7.75 kcal mol $^{-1}$.

Canonical Set-C endpoints have SD_{prop} 1.17–3.62 kcal mol $^{-1}$ and frame-level SEM 0.16–0.61 kcal mol $^{-1}$. Snapshots are temporally autocorrelated and each state has a single trajectory, so neither quantity quantifies between-replicate uncertainty. The 7.75 kcal mol $^{-1}$ K76A inter-replicate difference illustrates this limit. The WT replicate offset was 2.01 kcal mol $^{-1}$ (Table S18), used only as a within-protocol endpoint diagnostic, not applied to MD distance ratios.

Interaction fingerprints (ProLIF 2.2.1, 100 frames per trajectory, sixteen systems) identify a sparse set of persistent contacts: four to eleven contacts per system exceed 50 % occupancy, most at 100 %. Recurring anchors include PfCRT TYR16 (fully occupied in five of six PfCRT systems) and PfDHFR LEU46/MET55, consistent with the canonical binding pockets. PP-15 was docked to wild-type PfCRT and PfDHFR (Vina 1.2.7) under the Set-C protocol. Both systems completed equilibration and 10 ns production (Table S20) with bound fraction 1.000 (mean minimum distances 2.77 Å PfDHFR, 3.20 Å PfCRT). MM-GBSA (PBC-whole, GB^{OBC2} igb=5, 0.15 M salt): (−29.52 ± 0.33) kcal mol $^{-1}$ (PfDHFR) and (−27.79 ± 0.49) kcal mol $^{-1}$ (PfCRT; Table S20). No mutant or RRS estimate is reported; single-replicate endpoints are within-protocol feasibility contrasts.

Five-seed redocking was stable within 0.03 kcal mol $^{-1}$ on both receptors (Table S23). The same control was repeated for PP-01 on the canonical wild-type inputs (Table S23): dispersion within 0.05 kcal mol $^{-1}$ on both receptors. The canonical PfDHFR WT score (−7.50 kcal mol $^{-1}$) falls inside the five-seed spread (mean (−7.499 ± 0.021) kcal mol $^{-1}$). The canonical PfCRT WT score (−9.30 kcal mol $^{-1}$) lies 0.04 kcal mol $^{-1}$ above the observed range (−9.262–−9.225 kcal mol $^{-1}$) because that single run’s receptor-preparation detail is not preserved on disk; this offset is within the 0.05 kcal mol $^{-1}$ seed-dispersion bound, so the seed-stability conclusion is unchanged.

Table S20: *PP-15 single-ligand feasibility probe (outside the Set-C estimand). Vina scores are best-pose docking scores on wild-type receptors; MD columns report the canonical 10 ns production runs. MM-GBSA is GB^{OBC2} ($igb=5$), 0.15 M salt, mean over the frame window $\pm$ SEM (100 frames for PfDHFR, 80 for PfCRT; PBC-whole trajectories). No mutant or RRS estimate is reported.*

Target	Vina (kcal mol ⁻¹)	Protocol	MM-GBSA (kcal mol ⁻¹)	SEM	Bound fraction
PfDHFR (7F3Y)	-8.62	PASS; 10 ns complete	-29.52	0.33	1.000
PfCRT (6UKJ)	-7.81	PASS; 10 ns complete	-27.79	0.49	1.000

Table S21: *Lightweight robustness and transfer audits for the selected Set-C cohort. These post-selection diagnostics assess sensitivity of the computational ranking and class definitions; they do not establish biochemical affinity, biological resistance, target engagement, or mechanism of action.*

Audit	Result	Interpretation boundary
Leave-one-candidate-out	Complete two-target $n = 12$; PNS-RRS ρ ranged from -0.3182 to 0.0273 ; ACSI-RRS ρ ranged from -0.7636 to -0.3182	Rank associations are sensitive to candidate removal and remain exploratory
Target-stratified correlations	PfDHFR: PNS-RRS $\rho = -0.3497$; PfCRT: $\rho = -0.0417$	Target-specific patterns are not interchangeable with the pooled analysis
Score perturbation	$\pm 1 \text{ kcal mol}^{-1}$ perturbations generated 272 candidate-score records; class unchanged in 212/272 (77.9%)	Operational RRS classes show threshold sensitivity, especially to WT-score perturbation
Threshold sensitivity	The threshold grid used in the main analysis was retained unchanged in the source record	Thresholds are triage rules, not experimentally calibrated categories
Upstream activity-transfer record	The public ChEMBL panel was checked against its source record; it contains general activity labels but no paired PfDHFR/PfCRT WT-mutant measurements	The record is contextual and cannot serve as an independent P2 replication or experimental validation
Additional docking sensitivity panel	39 ligands, 8 PfDHFR/PfCRT states, 312/312 finite records after declared EXT-039 embedding exclusion; bootstrap mean RRS 100.45 (99.59–101.32), Class-A fraction 0.974 (0.923–1.000)	Computational docking sensitivity analysis only; not experimental validation or an MD estimate
GNINA CNN rescoring	GNINA 1.3.2 rescored the same 312 poses using its CNN score (post-processing only); 312/312 scores were finite. Under the frozen RRS estimand, 38/38 eligible ligands were class A under both scoring layers, with 0 class discordances; mean RRS was 100.4 with Vina and 103.6 with CNN; ligand-level Spearman correlation was 0.558; per-mutant ρ : K76T 0.644, K76A 0.486, N51I 0.271, S108N 0.204, I164L 0.040, C59R -0.076	Consensus sensitivity: class-level retention is shared across scoring functions; per-mutant rank transfer is not supported (Vina-only remains canonical)
Set-C bootstrap CI95	$B = 1 \times 10^4$ resamples (seed 42) on the 17-candidate class table: class fractions A* 0.294 [0.118–0.529], A 0.059 [0.000–0.176], B 0.294 [0.118–0.529], C 0.294 [0.118–0.529], D 0.059 [0.000–0.176]. Per-mutant retention (WT = 100): PfCRT K76T 85.4 [81.0–90.4], K76A 87.0 [83.1–90.9] ($n = 17$, intervals exclude 100); PfDHFR N51I 74.7 [67.2–85.2], C59R 73.7 [67.4–82.1], S108N 75.3 [67.7–85.8], I164L 76.8 [68.9–88.2] ($n = 12$, intervals overlap 100 for I164L)	PfCRT retention-not-gain is robust to resampling; PfDHFR intervals are wide and do not support per-mutant rank claims
MD-filter retention gate (pilot scope)	Intersection of docking-RRS $\geq 80\%$ with pilot MD-RRS < 100 on 12 gate-eligible rows (PP-01/PP-02): 7/12 rows pass both gates. PP-01 satisfies the two-target gate (PfCRT via K76A 97.8; PfDHFR four of four mutants 91.0–98.9); PP-02 does not (PfDHFR rows lack a docking wild-type anchor; single-target gate). PP-01 PfCRT K76T (102.7) is the sole row above parity; this dimensionless distance result is not compared with the $2.0 \text{ kcal mol}^{-1}$ MM-GBSA endpoint offset	Pilot-scope consistency check; full-panel (17×8) MD-RRS is <code>NOT_COMPUTED</code> ; the gate does not replace docking-RRS and is not a biological resistance claim

Table S22: *Upstream workflow context used to define the Set-C cohort. The upstream library and parts of the docking provenance overlap with the present analysis; these records therefore provide provenance and methodological context, not an independent validation set.*

Contextual record	Use in this study	Interpretation boundary
Redocking and enrichment benchmarks	Document the fixed grid protocol and general pose-reproduction behaviour	Do not validate mutant-state score retention or RRS
Physicochemical and drug-likeness profiles	Describe the source library and the filtered PP-01–PP-17 cohort; 17/17 SMILES identities were reconciled	Shared candidate identity and workflow provenance preclude independent-replication claims; properties are not activity measurements
Chemical-space and scaffold analyses	Describe the origin and structural diversity of the source library	Set-C is a filtered prioritization cohort, not a representative sample of the source library
MPO sensitivity	Document the selection procedure and its local sensitivity	Does not test RRS, trajectory geometry, or network-weighted ranking
Shared candidates and workflow provenance	Explicitly define the relationship between the upstream record and the present Set-C analysis	The present study is not an independent validation of the upstream workflow

Table S23: *Multi-seed redocking reproducibility for candidates PP-15 and PP-01 (AutoDock Vina [3], five random seeds; lowest-energy mode). Values in kcal mol^{−1}. Each control reuses the exact canonical ligand preparation and grid box that produced the candidate’s canonical single-run score. For PP-01 PfCRT WT, the canonical value (−9.300) lies 0.04 kcal mol^{−1} above the observed seed range because the original receptor-preparation detail of that single run is not preserved on disk; this offset is within the 0.05 kcal mol^{−1} protocol-local seed-dispersion bound. Seed-sensitivity controls measure docking-stochasticity reproducibility, not binding or resistance.*

Candidate–Target	Mean	Range	Canonical run	Seed sensitivity
PP-15 PfDHFR WT	−8.590 ± 0.007	−8.603/ − 8.585	−8.621	≤ 0.03 kcal mol ^{−1}
PP-15 PfCRT WT	−7.822 ± 0.011	−7.840/ − 7.810	−7.810	≤ 0.03 kcal mol ^{−1}
PP-01 PfDHFR WT	−7.499 ± 0.021	−7.529/ − 7.475	−7.500	≤ 0.05 kcal mol ^{−1}
PP-01 PfCRT WT	−9.250 ± 0.015	−9.262/ − 9.225	−9.300	≤ 0.05 kcal mol ^{−1}

The versioned machine-readable outputs (long-format and per-candidate threshold-margin tables) are archived with the source-data records. The input SHA-256 and analysis parameters are recorded in the accompanying manifest.

Table S24: Descriptive margins of the available-target RRS observations around the operational 70 % and 80 % thresholds; margins are in percentage points (pp). A candidate is flagged when at least one finite observation lies within 5 pp of either threshold. The analysis used the frozen canonical classification table and did not modify class assignments; it is sensitivity information rather than calibration against experimental resistance.

Candidate	Finite RRS obs.	Minimum abs. margin (pp)	Near-threshold obs.	Flagged
PP-01	12	3.87	2	Yes
PP-02	4	7.91	0	No
PP-03	12	0.00	12	Yes
PP-04	12	1.25	8	Yes
PP-05	4	0.67	4	Yes
PP-06	4	10.61	0	No
PP-07	12	1.84	12	Yes
PP-08	12	2.76	8	Yes
PP-09	12	0.56	12	Yes
PP-10	12	3.64	8	Yes
PP-11	4	2.50	4	Yes
PP-12	12	0.38	8	Yes
PP-13	4	19.40	0	No
PP-14	12	1.20	8	Yes
PP-15	12	6.24	0	No
PP-16	12	0.11	12	Yes
PP-17	12	3.24	2	Yes

Table S25: Complete eight-system explicit-solvent molecular dynamics (10 ns) and MM-GBSA trajectory audit for the Class-A* lead candidate **PP-01** across wild-type and resistant mutant states of PfDHFR and PfCRT. All eight systems passed the geometric trajectory-QC contract (bound fraction = 1.000, mean minimum heavy-atom distance < 3.10 Å); this is pose retention, not a resilience proof. MD distance RRS (MD-RRS_d) measures mean minimum heavy-atom distance retention relative to the corresponding wild-type anchor (> 100 % indicates looser binding). MM-GBSA binding free energies (ΔG_{MMGBSA}) are computed over 100 snapshots (mean $\pm$ SD_{prop}). The PP-01 PfCRT K76A canonical endpoint is displayed with a 7.75 kcal mol⁻¹ inter-replicate sensitivity diagnostic (see K76A replicate table); no endpoint resilience claim is made for K76A.

Target	System State	Mean Min Dist (Å)	ΔG_{MMGBSA} (kcal mol ⁻¹)	SD _{prop}	MMG-RRS (%)	MD-RRS _d (%)
PfDHFR	Wild-Type (WT)	2.94	-25.29	2.94	100.0	100.0
PfDHFR	N51I	2.67	-29.83	1.50	118.0	91.0
PfDHFR	C59R	2.91	-26.56	1.91	105.0	98.9
PfDHFR	S108N	2.82	-30.21	3.62	119.5	96.1
PfDHFR	I164L	2.85	-26.13	2.34	103.3	97.2
PfCRT	Wild-Type (WT)	2.94	-30.61	1.65	100.0	100.0
PfCRT	K76T	3.02	-29.51	2.65	96.4	102.8
PfCRT	K76A	2.88	-35.29	1.17	115.3	97.8

Table S26: Retrospective confrontation of static Vina $\Delta\Delta G$ with published ChEMBL genotype-resolved PfDHFR IC_{50} fold-shifts (target CHEMBL1939; 28 compounds). IC_{50} values are whole-parasite growth measurements in DHFR-genotyped, non-isogenic strains (wild-type TM4/8.2; double C59R/S108N K1CB1; triple Csl-2; quadruple N51I/C59R/S108N/I164L VI/S) and are used as a descriptive phenotype-level reference, not as binding-affinity ground truth. $\Delta\Delta G = s_{\text{mutant}} - s_{\text{WT}}$ (kcal mol⁻¹; positive = predicted resistance). Agreement: 1 = predicted and observed direction match, 0 = mismatch. Descriptive only; this is not experimental validation of the Set-C RRS classes. Source: ChEMBL (target CHEMBL1939) curation and the Vina protocol described in Methods.

Compound	IC ₅₀ (nM) / fold					Vina $\Delta\Delta G$ (kcal mol ⁻¹)			Agr.
	WT	Double	Triple	Quad	Quad/WT	Double	Triple	Quad	
CHEMBL158570	3900	2600	3100	2000	0.5	-0.57	-0.65	-0.64	111
CHEMBL158639	180	180	290	280	1.6	-0.20	-0.27	-0.25	100
CHEMBL159052	3000	7400	5200	3200	1.1	-0.19	-0.21	-0.23	000
CHEMBL161009	60	480	490	510	8.5	-0.12	-0.20	-0.21	000
CHEMBL161034	120	3100	4800	3200	26.7	0.14	0.13	0.13	111
CHEMBL161117	2200	2700	3400	3500	1.6	-0.27	-0.27	-0.29	000
CHEMBL161451	650	3100	3300	740	1.1	0.13	0.10	0.07	111
CHEMBL161673	20	2400	4100	3300	165.0	-0.14	-0.19	-0.19	000
CHEMBL161757	5500	3100	3900	3400	0.6	-0.89	-1.00	-0.99	111
CHEMBL21921	670	3000	3600	4300	6.4	-0.00	-0.10	-0.11	000
CHEMBL22403	350	3300	2900	4500	12.9	0.00	-0.11	-0.65	100
CHEMBL22693	210	2300	2700	2700	12.9	-0.23	-0.30	-0.31	000
CHEMBL278813	410	2700	2700	4000	9.8	0.20	0.19	0.16	111
CHEMBL278847	2.7	1300	2500	3900	1444.4	-0.15	-0.21	-0.25	000
CHEMBL329687	560	310	250	200	0.4	-0.11	-0.12	-0.15	111
CHEMBL346932	2100	–	1900	2000	1.0	0.09	0.02	0.01	-00
CHEMBL346963	40	260	140	330	8.2	-0.59	-0.68	-0.59	000
CHEMBL346982	250	1000	2900	380	1.5	0.06	0.07	0.05	111
CHEMBL348151	3200	2800	3400	2800	0.9	0.24	0.21	0.21	010
CHEMBL352195	2800	3300	4000	3800	1.4	-0.17	-0.12	-0.12	000
CHEMBL352441	1900	–	1900	4100	2.2	-0.07	-0.06	-0.13	-10
CHEMBL354011	4000	4900	7500	3300	0.8	-0.08	-0.01	-0.06	001
PYRIMETHAMINE	80	30900	41700	73500	918.8	-0.12	-0.27	-0.24	000
CHEMBL416552	60	1300	3000	2300	38.3	0.02	-0.03	-0.02	100
CHEMBL422005	1100	–	1900	4500	4.1	0.02	-0.08	-0.06	-00
CHEMBL444605	1600	–	1800	2100	1.3	-0.03	-0.09	-0.10	-00
CHEMBL7130	50	220	520	220	4.4	-0.16	-0.12	-0.16	000
CYCLOGUANIL	40	2400	37900	5800	145.0	-0.22	-0.32	-0.30	000

Use of Artificial Intelligence

During the preparation of this manuscript in September 2026, the authors utilized AI-based coding assistants to support code review, provide structured text suggestions, and perform consistency checks. The authors designed all analyses, maintained full control over the scientific interpretation, and verified all computational results and quantitative claims against the original source records. The tools do not hold authorship; the authors remain fully accountable for all content in this supporting information.

Acknowledgements

No AI tool was used to generate unverified scientific content for this supporting information. All quantitative claims were verified by the authors against the source records.

References

- [1] Myke Vital Sao Temgoua, Jean-Pierre Tchapel Njafa, Serge Guy Nana Engo, Penabei Samafou, and Wilfred Fon Mbacham. Topological and tensor-network representations resolve chemical paradoxes in African antimalarial natural products. *ChemRxiv*, 2026. doi: 10.26434/chemrxiv.15007167/v1. Preprint.
- [2] Amr Alhossary, Stephanus Daniel Handoko, Yuguang Mu, and Chee-Keong Kwoh. Fast, accurate, and reliable molecular docking with QuickVina 2. *Bioinformatics*, 31(13):2214–2216, 2015. doi: 10.1093/bioinformatics/btv082.
- [3] Jerome Eberhardt et al. AutoDock Vina 1.2.0: new docking methods, expanded force field, and python bindings. *Journal of Chemical Information and Modeling*, 61:3891–3898, 2021. doi: 10.1021/acs.jcim.1c00203.
- [4] Damian Szklarczyk et al. STRING v13: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Research*, 51:D633–D641, 2023.
- [5] Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. DiffDock: diffusion steps, twists, and turns for molecular docking. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=kKF8_K-mBbS. arXiv:2210.01776.
- [6] Gemma Turon, Joseph Hlozek, James G. Woodland, et al. First fully-automated AI/ML virtual screening cascade implemented at a drug discovery centre in Africa. *Nature Communications*, 14:736, 2023. doi: 10.1038/s41467-023-41512-2.
- [7] Kyle Swanson et al. ADMET-AI: a machine learning ADMET platform for evaluation of large-scale chemical libraries. *Bioinformatics*, 40:btae416, 2024. doi: 10.1093/bioinformatics/btae416.
- [8] PlasmoDB Consortium. PlasmoDB: The *Plasmodium* genomic resource. <https://plasmodb.org/>, 2026.
- [9] Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.

- [10] M. S. Valdés-Tresanco, M. E. Valdés-Tresanco, P. A. Valiente, and E. Moreno. gmx_MMPBSA: A new tool to perform end-state free energy calculations with GRO-MACS. *Journal of Chemical Theory and Computation*, 17(10):6281–6291, 2021. doi: 10.1021/acs.jctc.1c00645.
- [11] J. Huang, S. Rauscher, G. Nawrocki, T. Ran, M. Feig, B. L. de Groot, H. Grubmüller, and A. D. MacKerell. CHARMM36m: an improved force field for folded and intrinsically disordered proteins. *Nature Methods*, 14(1):71–73, 2017. doi: 10.1038/nmeth.4067.